

Converting Written Health-Care Policy into Executable, Auditable Rules

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The Concept

Health care runs on written policy: billing and coding manuals, clinical practice guidelines, and payer-provider contracts. These documents are authored in natural language for human readers, yet the payment and care decisions that depend on them must execute at machine scale consistently, and defensibly. Converting written policy into rules, features, or model inputs is the discipline of transforming that prose into executable artifacts without losing the traceability a regulator or an appeals reviewer demands.

The canonical example in payment integrity is the **CMS National Correct Coding Initiative (NCCI)**, whose Procedure-to-Procedure (PTP) edits and accompanying multi-hundred-page Policy Manual govern which code pairs may be billed together (Centers for Medicare & Medicaid Services [CMS], n.d.-a, n.d.-b). Today, turning each quarterly manual update into running edit logic is largely manual analyst work which is slow, costly, and inconsistent.

The recent advance is that large language models (LLMs) can read policy prose and draft structured rule candidates. The critical design constraint is **the seam**: the LLM may convert language into a draft, but it must never decide a payment. Decisions stay on a deterministic engine whose every output cites the rule and rule-set version that produced it. Between the two sits a mandatory human gate where a reviewer may approve, **correct a field**, or reject a draft before it is ever stored. This separation keeps the speed of LLMs while preserving the auditability adjudication requires.

Trends

Three trends converge on this problem. First, retrieval-augmented generation (RAG) and structured extraction let models ground their outputs in source documents and emit typed objects rather than free text (Lewis et al., 2020), making “policy → rule candidate” tractable. Second, agentic, human-in-the-loop workflows are replacing one-shot prompting: a pipeline retrieves the relevant manual section, extracts a candidate, routes it to a human gate for approval or correction, and only then commits it to a versioned store. Third, regulatory and operational pressure for transparency is rising i.e payers must explain denials, and unexplained or model-driven denials invite appeals and scrutiny. Together these reward architectures that are fast at conversion but conservative and explainable at decision time.

The countervailing reality is that LLMs hallucinate and cannot serve as a system of record (Ji et al., 2023). The mature pattern is therefore not “LLM decides” but “LLM drafts, a human approves or corrects, and a deterministic engine decides”, precisely the boundary payment integrity needs.

Opportunities and Threats

Opportunities. Administrative complexity is the single largest category of waste in U.S. health care. Total waste is estimated at \$760–935 billion annually (Shrank et al., 2019). Faster, cheaper, more consistent conversion of policy into executable edits attacks this directly: shorter turnaround on quarterly NCCI updates, fewer missed edits, and a defensible audit trail for every decision. The same pipeline generalizes beyond NCCI to clinical guidelines and contract terms, and the convert-vs-decide seam is a reusable, productizable architecture aligned with Cotiviti’s payment-integrity mission (Cotiviti, n.d.).

Threats. The principal risk is misplaced trust, letting a model’s output reach a payment decision, a compliance and reputational hazard. Secondary risks include:

- silent drift between policy versions;
- ungrounded (“hallucinated”) rule candidates;
- intellectual-property constraints on code sets (CPT is AMA-licensed; American Medical Association [AMA], n.d.);
- extraction errors that are internally inconsistent — for example, a candidate whose structured modifier indicator contradicts its own rationale; and
- the temptation to inflate extraction-fidelity metrics instead of reporting an honest, human-gated number.

A demonstrator that hides the human gate, the corrections it makes, or the citations would defeat the purpose.

Strategic Recommendations for Cotiviti

1. **Adopt the seam as policy.** Standardize on an architecture in which LLMs convert policy to draft rules and a deterministic engine adjudicates, with every disposition citing a rule ID and ruleset version. This is auditable by construction and aligns with Cotiviti’s payment-integrity mission.
2. **Invest in a human-gated rule factory.** Build or productize a pipeline — retrieve → extract → human approve/correct → versioned store → adjudicate. That compresses the quarterly policy-to-edit cycle while keeping a mandatory human checkpoint able to correct a field, not merely approve or reject it.
3. **Measure honesty, not vanity.** Track extraction fidelity separately from adjudication correctness. Report extraction recall over the extractable subset, the pairs actually stated as code pairs in the prose and treat an imperfect, human-gated number as the expected, defensible result.
4. **Generalize deliberately.** Once proven on NCCI, extend the same convert-vs-decide boundary to clinical guidelines and payer–provider contract terms, where the same separation applies.

The Proof of Concept: PolicyForge

PolicyForge implements the recommended seam end-to-end. It ingests the CMS NCCI Policy Manual and PTP table, retrieves relevant chapters (with a direct-vs-vector retrieval ablation), and uses an LLM to extract RuleCandidate drafts grounded to a verbatim source quote. Each candidate is routed through a human-gate interrupt where a reviewer can approve it, **correct a field such as the modifier indicator**, or reject it. Approved rules are persisted with full provenance including fields approver, timestamp, source quote, and grounding flag in a versioned store, and synthetic claims are adjudicated on a deterministic engine that cites the rule and rule-set version for every deny or flag. No model, embedding, or randomness ever touches the decision path. The seam is enforced in code and verified by an automated test suite of 100+ tests.

Evaluation is reported in two tracks, deliberately kept separate. **Track A** measures extraction fidelity as recall over the extractable subset; recall over the full table is low by construction, because the manual states most guidance as principles rather than explicit code pairs. **Track B** measures adjudication correctness with deterministic exact-match scoring and is one of the main pillars of the pipeline. The gate's ability to correct a wrong indicator before storage is why an imperfect Track A number, honestly reported, is the credible result: the demo's purpose is to show the seam, the human gate, and the provenance honestly, not to manufacture a perfect score.

References

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